

that we see heads is 50 percent. If we flip a fair coin 10 times, we expect to see 5 heads and 5 tails, but it's possible that we will see 8 out of 10 heads, or 80 percent heads. The law of large numbers holds that if we perform the same experiment a large number of times, the average of the results should be closer to the expected value—50 percent heads—and will become closer still as more trials are performed. So, if we flip the coin 100 times more, the average number of heads for the whole 110 flips will be closer to 50 percent simply because the most likely outcome from 100 additional flips is 50 heads. If we add the 50 heads from the second trial to the eight heads from the first trial, the proportion of heads for the whole series declines from $8 \div 10$ or 80 percent, to $8 \div 10 + 50 \div 100 = 58 \div 110$ or 52.7 percent. If we flipped the coin 1,000 times more, we expect to observe 500 heads and 500 tails, and the proportion of heads for the whole series would decline from $58 \div 110$ to $558 \div 1110$ or 50.5 percent. The law of large numbers holds that as we continue to flip the coin, it is highly likely that the proportion of heads we observe will converge toward 50 percent.

It is important to note that the observed proportion of heads does not converge toward the expected 50 percent because a run of tails follows the run of heads. It occurs because the subsequent ratio of heads and tails conforms to the underlying probability (50 percent heads, 50 percent tails), and the absolute error (8 heads) becomes proportionately smaller relative to the number of trials. The belief in a run of tails following a series of heads is known as the *gambler's fallacy*. This is the mistaken idea that deviations from the expected probabilities in independent trials of a random process make future deviations in the opposite direction more likely. In other words, this is the belief that a tail is due following a run of heads. Coin flips are independent trials of a random process, which means that the likelihood of a head or tail is independent of any earlier appearance of a head or a tail. Even after the run of eight heads, given a fair coin, the chance of a head remains at 50 percent.

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Mean reversion pervades finance, but it's not the purely statistical mean reversion of the coin flip. Mean reversion in the markets looks a lot more like the gambler's fallacy made real—movements in security prices, individually and in aggregate, tend to be followed by subsequent price movements in the opposite direction. The more extreme the initial price movement, the greater will be the subsequent adjustment in the opposite direction. This phenomenon is observable in the financial results of businesses, in the market prices of securities, in the movements of indices, and in the results of investment